

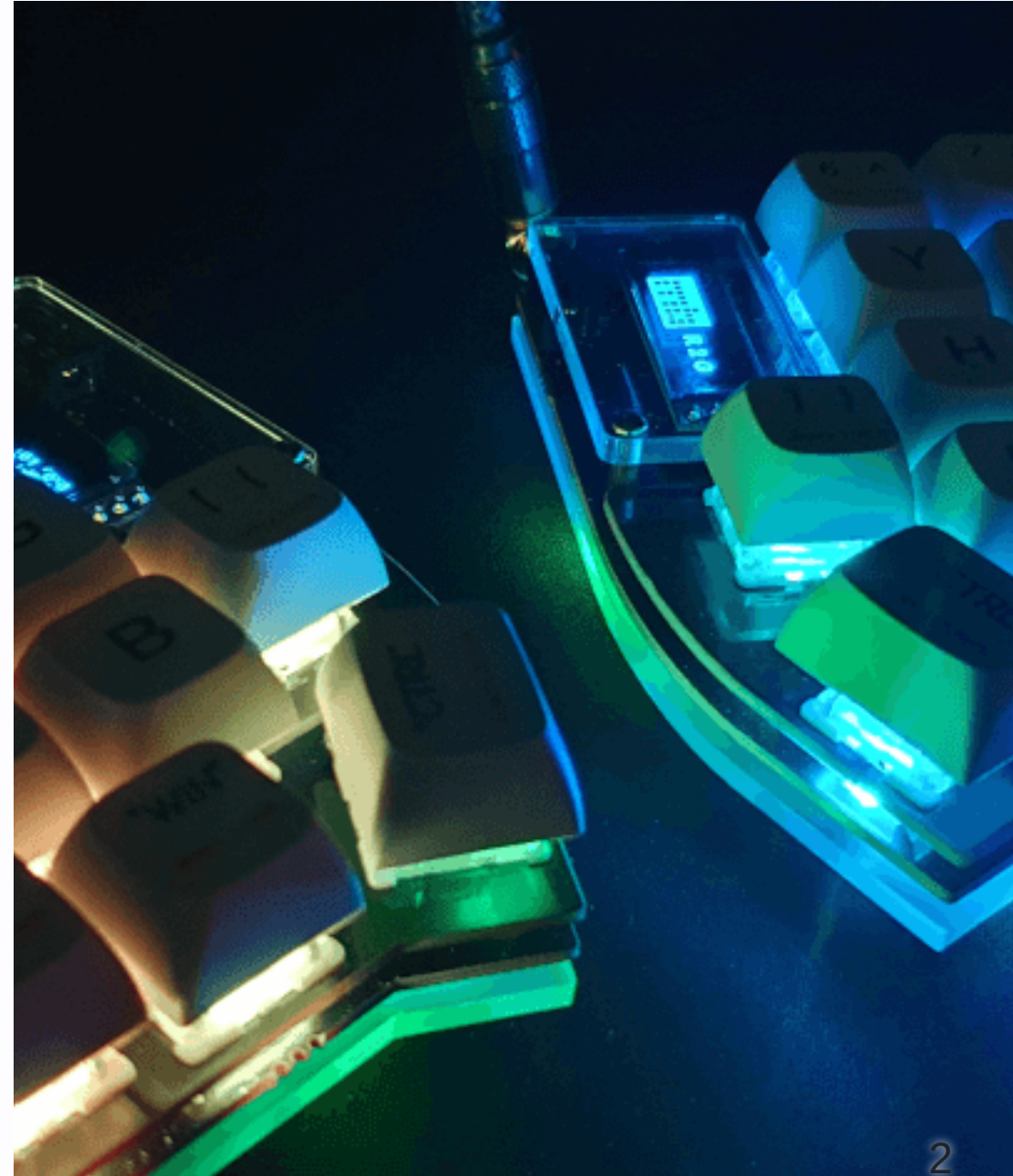
Using HTMX To Make Interactive Elements In Drupal

DrupalCamp England 2026

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Source Code

- Talk is available at:
<https://github.com/hashbangcode/drupal-hmx-talk> or via QR code
- All code seen can be found at:
 - <https://bit.ly/46arnzj> HTMX examples
 - <https://bit.ly/3OHAIIR> Drupal examples
- More resources at
www.hashbangcode.com



Using HTMX To Make Interactive Elements In Drupal

HTMX In Drupal

- HTMX was added to Drupal in 11.3 as a core component.
- This is the new standard for ajax requests.
- All existing ajax features will be re-written in HTMX.

HTMX

What Is HTMX?

- JavaScript framework.
- Allows ajax calls and CSS Transitions *without writing JavaScript code*.
- Small at 16K (minified and gzipped).
- No other dependencies.
- Platform agnostic.
- Plugins extend functionality.

What Is HTMX?

- Behaviour is added to HTML attributes.
- Any element can issue a web request.
- All responses should be in HTML.

Installing HTMX

Download and include the single JavaScript file.

```
<script src="htmx.min.js"></script>
```

There are CDN options available.

We are now ready to use HTMX.

HTMX Attributes

- Prefixed with `hx-` or `data-hx-`.
- 11 attributes are commonly used.
- There are another 20-30 that add more functionality.

HTMX Attributes - Example

```
<button hx-get="index.php" hx-target="#div1">Submit</button>  
  
<div id="div1"></div>
```

This example will:

- `hx-get` = Send a "get" request to the index.php path.
- `hx-target` = Write the response in the element with the ID "div1".

HTMX Requests

- All requests from HTMX will have the `hx-request` header.
- Certain attributes will add further headers to the request.

Responding To The Request

Your response should be in plain HTML.

We could respond to the previous request with:

```
<p>Button clicked!</p>
```

Which would be injected into our page like this.

```
<div id="div1"><p>Button clicked!</p></div>
```

Responder Class In PHP

A simple class to respond to HTMX requests.

```
class Htmx {  
    public static function isHtmxRequest(): bool {  
        return isset($_SERVER['HTTP_HX_REQUEST']) && $_SERVER['HTTP_HX_REQUEST'] === 'true';  
    }  
  
    public static function isGet(): bool {  
        return $_SERVER['REQUEST_METHOD'] === 'GET';  
    }  
}
```

Responding To The Request

We detect the `hx-request` header and the correct HTTP method before responding.

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo '<p>Button clicked at ' . date('r') . ' .</p>';  
}
```

The div now looks like this:

```
<div id="div1"><p>Button clicked at Sun, 08 Feb 2026 16:00:23 +0000.</p></div>
```

HTMX Common Attributes

HTTP Verbs

Different HTTP verbs are available through attributes.

- `hx-get` does a "get" request.
- `hx-post` does a "post" request.
- `hx-delete` does a "delete" request.
- `hx-patch` does a "patch" request.
- `hx-put` does a "put" request.

hx-target

The target element to be swapped.

```
<button hx-get="index.php" hx-target="#div1">Submit</button>  
  
<div id="div1"></div>
```

hx-trigger

Defines the action that will trigger the request.

```
<div hx-post="index.php" hx-trigger="click">Click me</div>
```

- Can be left out for elements that have default triggers.
- Eg. `<button>` elements automatically have a `click` trigger.

hx-trigger Examples

```
<div hx-post="index.php" hx-trigger="click">Click</div>
```

```
<div hx-post="index.php" hx-trigger="click once">Click</div>
```

```
<div hx-post="index.php" hx-trigger="load, click">Load & Click</div>
```

```
<div hx-post="index.php" hx-trigger="revealed"></div>
```

```
<div hx-post="index.php" hx-trigger="load delay:500ms"></div>
```

```
<input hx-get="index.php" hx-trigger="keyup delay:1s" />
```

hx-select

Select content to swap with the target element from the response.

HTMX:

```
<button hx-get="index.php" hx-select="#response" hx-target="#div1">Click</button>  
<div id="div1"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo '<p id="response">Button clicked at ' . date('r') . ' .</p>';  
    echo '<p>Some extra content that won\'t get displayed.</p>';  
}
```

hx-select-oob

Select an "out of band" element to also target in the response.

HTMX:

```
<button hx-get="index.php" hx-select-oob="#div1">Send Request</button>  
<div id="div1"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo 'Request Sent';  
    echo '<div id="div1">Button clicked at ' . date('r') . ' .</div>';  
}
```

hx-select-oob - Continued

You can comma separate this property to do multiple things.

HTMX:

```
<button hx-get="index.php" hx-select-oob="#div1,#div2">Send Request</button>  
<div id="div1"></div> <div id="div2"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo 'Request Sent';  
    echo '<div id="div1">Button clicked at:</div>';  
    echo '<div id="div2">' . date('r') . '</div>';  
}
```

hx-swap

- Controls how an element will be swapped into the page.
- Defaults to `innerHTML` - replace the HTML contents.
 - `outerHTML` - Replace the element.
 - `textContent` - Replace contents without parsing as HTML.

hx-swap - Continued

- `beforebegin` , `afterbegin` , `beforeend` , `afterend` - Place the response before or after the element and its children.
- `delete` - Delete the target element from the page.
- `none` - Do nothing with the response (but still process out of band items).

hx-swap-oob

Swap an "out of band" element. Similar to `hx-select-oob` but the response tells HTMX what to swap.

HTMX:

```
<button hx-get="index.php" hx-swap="none">Send Request</button>  
<div id="div1"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo '<div id="div1" hx-swap-oob="true">Button clicked at ' . date('r') . '</div>';  
}
```

CSS Transitions

- HTMX adds classes to elements during different events.
- `htmx-added` is added to new content before it is added to the page.
- We can use these classes to animate what HTMX is doing with CSS transitions.

CSS Transitions

The following will fade in a bit of content inside the `div1` element as it is loaded into the page.

```
#div1 p {  
    opacity: 1;  
    transition: opacity 1s ease-in;  
}  
#div1 p.htmx-added {  
    opacity: 0;  
}
```

Configuring HTMX

- It is possible to configure HTMX via a number of settings.
- Can be set through meta tags (preferred)

```
<meta name="htmx-config" content='{ "defaultSwapStyle": "outerHTML" }'>
```

Or JavaScript.

```
<script>  
htmx.config.defaultSwapStyle = 'outerHTML';  
</script>
```

DEMO!

HTMX In Drupal

HTMX In Drupal

- HTMX is included in Drupal as a library.
- A few classes exist to assist with adding attributes and responding to requests.

HTMX in Drupal

The standard usage of HTMX in Drupal is to use the `data-hx-` prefix for attributes.

Eg:

```
<button data-hx-get="index.php" data-hx-target="#div1">Submit</button>  
  
<div id="div1"></div>
```

Drupal Libraries

HTMX can be included into the page through a Drupal library.

- `core/htmx` - The core HTMX library.
- `core/drupal.htmx` - Additional scripts for Drupal.
- You rarely need to inject these libraries directly.

Drupal Libraries

The library `core/drupal.htmx` includes three files:

- `htmx-assets.js` - Adds assets the current page requires.
- `htmx-behaviors.js` - Connect `Drupal.behaviors` to HTMX inserted content.
- `htmx-utils.js` - Helper functions for the other two files.

Drupal Integration

- The `\Drupal\Core\Htmx` class is used as a wrapper around the HTMX libraries and attribute injection for render arrays.
It can also inject headers for responding to HTMX.
- `\Drupal\Core\Htmx\HtmxRequestInfoTrait` is used to detect HTMX requests and associated headers.

The Htmx Class

- Has a number of methods that set up attributes.

```
$output['element'] = [  
    '#type' => 'html_tag',  
    '#tag' => 'p',  
    '#value' => 'Content'  
];  
  
$htmx = new Htmx();  
$htmx->get()  
    ->swap('afterend')  
    ->trigger('revealed')  
    ->applyTo($output['element']);
```

```
<p  
    data-hx-get="/route"  
    data-hx-swap="afterend"  
    data-hx-trigger="revealed">  
Content  
</p>
```

The Htmx Class

- The Htmx class can also be used with a different syntax.

```
(new Htmx())  
  ->get()  
  ->swap('afterend')  
  ->trigger('revealed')  
  ->applyTo($output['element']);
```

HTMX Drupal Response

- Drupal has the ability to respond to a HTMX response with HTML.
- This uses the class `Drupal\Core\Render\MainContent\HtmxRenderer`.
- Renders a little HTML document and sends this upstream.

HTMX Drupal Response

You can respond with using the `HtmxRenderer` service, but it is often easier to invoke it.

`HtmxRenderer` is invoked by:

- `_wrapper_format=drupal_htmx` as a query on the incoming request.
- The `_htmx_route=true` on the route replying to the request.

HTMX With Controllers

HTMX With Controllers

- Think about your approach.
 - One route: with logic to separate out the HTMX response.
 - Two routes: one for rendering the page, one marked with `_htmx_route: TRUE` for the HTMX response.

HTMX With Controllers

To use the `HtmxRequestInfoTrait` trait you need to add the request stack service.

```
namespace Drupal\mymodule\Controller;

use Drupal\Core\Controller\ControllerBase;
use Drupal\Core\Htmx\HtmxRequestInfoTrait;

class MyController extends ControllerBase {

    use HtmxRequestInfoTrait;

    public function __construct(protected RequestStack $requestStack) {}

    protected function getRequest() {
        return $this->requestStack->getCurrentRequest();
    }
}
```

HTMX With Controllers

Then, in your action:

```
public function action() {  
    if ($this->isHtmxRequest()) {  
        // Respond to HTMX request.  
    }  
  
    // Respond to normal controller action.  
    // Generate markup to set up the HTMX request.  
}
```

HTMX With Controllers

The `__htmx_route: TRUE` is used to create the HTMX response action.

```
mymodule_controller_action_htmx:  
  path: '/htmx-response'  
  defaults:  
    _controller: '\Drupal\mymodule\Controller\MyModuleController::htmx'  
  requirements:  
    _permission: 'access content'  
  options:  
    _htmx_route: TRUE
```

HTMX With Forms

HTMX With Forms

- Forms work in much the same way as Controllers.
- You need to decorate the elements inside the `buildForm()` method.
- There is no need to create callback functions, you pull the form elements out of the response.

HTMX With Forms

Decorate the form elements using the Htmx class.

```
public function buildForm(array $form, FormStateInterface $form_state) {  
    $form['text'] = [  
        '#type' => 'textfield',  
        '#title' => $this->t('Text'),  
    ];  
  
    (new Htmx())  
        ->post()  
        ->target('*:has(>input[name="text"])')  
        ->select('*:has(>input[name="text"])')  
        ->trigger('keyup delay:1s')  
        ->applyTo($form['text']);  
  
    return $form;  
}
```

HTMX With Forms

- `HtmxRequestInfoTrait` is part of the `FormBase` class.
- The `request_stack` service is also part of the form so there is no need to inject this.

```
public function buildForm(array $form, FormStateInterface $form_state) {  
    if ($this->isHtmxRequest()) {  
        // React to HTMX request.  
    }  
    // Rest of the form.  
}
```

HTMX With Forms

- The HTMX response from forms is the entire form, so you *need* to add a `hx-select` (or similar) to pick out the relevant part of the response.
- Forms need to be consistent. You can't just throw elements into the form markup as the elements need to exist in the form build.

DEMO!

Drupal Needs You!

- HTMX is in Drupal 11.3 as a fully featured system.
- All of the ajax subsystem now needs to be converted to HTMX.
- There's a lot of work to be done.
- <https://www.drupal.org/community-initiatives/replace-ajax-api-with-htmx>

HTMX Module

- <https://www.drupal.org/project/htmx>
- Some extra tools and examples of HTMX in a Drupal setting.
- Has a Views plugin to display Views as HTMX.
- Has a HTMX debugger that logs all HTMX events in the browser console.

Resources

- #htmx on Drupal Slack.
- htmx.org - <https://htmx.org>
- [HTMX Labs](https://htmxlabs.com) - <https://htmxlabs.com>

Questions?

- Slides: <https://bit.ly/4tFKUBK>



Thanks!

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